Lodestar: bankable demand forecasts for mini-grids

What P90 energy-yield reports did for solar project finance, Lodestar does for mini-grid demand.

Team Lodestar · Tushar Agarwal · Brief 03 "Predict mini-grid demand before building" · Track 2, Digital Solutions & AI · Setting B (non-residential: developers, lenders, RBF programmes) · Code: github.com/TusharTechs/lodestar · Live app: lodestar-utfz.onrender.com · Demo video: youtu.be/lSxJWJSjBLc

Community & target user	**User:** mini-grid developers planning sites in unelectrified villages (first: Nigeria, Kenya; validated in DRC). **Payers:** the lenders, DFIs and results-based-financing (RBF) programmes that fund them. **Who benefits:** households and businesses in the 217,000 mini-grids needed to serve 490 million people by 2030 (ESMAP, $127B).
The problem	Developers size systems on demand guesses. Survey-based estimates miss individual customers by **65%** (CrossBoundary Innovation Lab). Oversizing a 30 kW site by 100% drops IRR from 12.5% to 6.1%; undersizing causes evening brownouts. **Root cause:** no forecast is (a) built from data that exists before construction, (b) honest about its range, (c) shows how demand *grows*, and (d) is ever checked against meters. Lenders cannot price what they cannot bound.
Proposed solution	Click any village. Lodestar estimates potential customers from building footprints and OpenStreetMap, then forecasts connections × consumption per connection month by month (growth shapes fitted on metered mini-grids, levels from AMDA 2024 benchmarks), with **P90/P50/P10** from 20,000 Monte Carlo draws. It explains the drivers in plain language, produces a load profile, system size and 8760-hour HOMER file, and runs a **lender stress test across every draw**: grant needed, debt that is safe on P90 demand, breach risk if debt is sized on P50. A deterministic one-click lender memo; no language model generates any number.
Proof of concept (built this weekend)	Working web app and API with 20+ villages in Nigeria, Kenya and DRC, live forecasts for any clicked location, a **Demand Bureau** upload that narrows the band with an operator's metered months, and a public **forecast ledger** (each forecast hashed and timestamped before outcomes are known).
Brief success metric: error vs metered demand on held-out sites	The organisers had no partner meter data, so we found a public CC0 panel of three metered DRC mini-grids (Lunanga et al. 2026). Forecasting the two village sites from **open data only** (footprints + OSM, no census, no site meters; ratios and shapes taken from the other sites): **5 of 6** site-years of metered energy inside the P90–P10 band (target 80%) **81%** of metered months inside the band **−26% / −29%** year-3 (sizing-year) P50 error, Lubero / Mutwanga **40%** mean annual error, years 1–3 (ref.: 37% per-site ML, Allee et al.) **In decisions:** replaying the real metered demand, Lodestar's build-at-P50-then-expand plan cut year-3 capacity mismatch by **72%** vs a desk-estimate build (Mutwanga: desk **+37%** oversized, Lodestar +6%); right-sizing on P50 instead of P10 cuts the cost of power **~48%**. Honest limits: two sites; a census-based flat estimate scored 36% but gives no range; after 12 metered months a conflict shock made persistence better. All reported in the app.
Estimated cost & who pays	**Running cost (measured):** under 1 CPU-s and 6 KB per forecast; open data, no API keys. **Break-even:** ~200 paid reports/yr = 3.3% of the 6,116 sites AMDA members plan (team cost assumed). **Revenue (proposed):** single-site forecasts free (adoption); a paid **Bankable Demand Report** per site for developers and lenders (proposed ~$300, under 0.1% of a ~$585k mini-grid); portfolio-screening subscriptions for DFIs and RBF programmes. **Lowest-income users pay nothing:** they gain from right-sized systems (lower tariffs, fewer brownouts); with a Nigeria-NEP-style $600/connection grant the viable sites need $0.20–0.40/kWh, within AMDA's observed tariffs.
6–18 month deployment path	• **M0–3:** backtest on 2–3 operators' historical meters under the Demand Bureau protocol (site-month aggregates only). • **M3–6:** pre-register forecasts for 10 sites before commissioning in the public ledger. • **M6–12:** score them against meters; integrate with one RBF programme's site-validation workflow. • **M12–18:** 5 countries; paid reports to lenders. DTECH 2027 (March) as the MVP milestone.
How it scales	Footprints and OSM cover all of Africa (and South/Southeast Asia, Latin America). A new country needs only its regional benchmark; Demand Bureau data then calibrates it. Running cost falls per site as forecasts are pre-computed. Portfolio screener ranks every village for RBF programmes (20 in 15 CPU-s: 1 RBF-bankable, 4 need an anchor, 12 solar-home-system, 3 near grid). Reaches the field by USSD/SMS (one 160-char text) and a 12 KB offline pack. Open source and self-hostable.
Biggest risk & how we test it	**Thin metered data** limits calibration and point accuracy. Test early: the M0–3 operator backtest; the **Demand Bureau** (give-to-get, k-anonymous: ≥3 sites, ≥2 operators per published cell; federated parameter-sharing option; neutral steward such as AMDA) grows the evidence base, and the **forecast ledger** makes every miss visible.
Prior work disclosure	None. All code, models and analysis were built during the sprint (Oct 30 – Nov 1, 2026). Open-source libraries and public datasets (listed in the README) were used; an AI coding assistant was used during development.

Sources: Brief 03; ESMAP Mini Grids for Half a Billion People (2022); AMDA Benchmarking Africa's Minigrids 2024; CrossBoundary Innovation Lab; Allee et al.; Lunanga, Stoop, Verpoorten & Desbureaux (2026) doi:10.7910/DVN/KPDREF (CC0); VIDA/Google/Microsoft/OSM Open Buildings; OpenStreetMap; GridFinder; NASA POWER; RAMP SSA archetypes (Stevanato 2026).